

CRE SENTINEL

Commercial Real Estate Debt Early-Warning System Portfolio Credit Risk Report

Prepared by Caina Emmanuella Gaspard

July 2026

All loan-level data in this report is synthetic. See Data Dictionary for full disclosure.

1. Executive Summary

CRE Sentinel is a portfolio-surveillance system built to give a commercial real estate lender continuous, scored visibility into loan-level credit risk, in place of static point-in-time reviews. This report summarizes findings from a synthetic 1,000-loan portfolio built to test the system end to end: a risk-scoring model, an interactive stress-testing tool, and a Power BI-ready surveillance dashboard.

Portfolio snapshot at baseline:

- Assets under management (current balance): \$14,653M
- Weighted average DSCR: 1.27x
- Weighted average LTV: 73.5%
- Average model risk score: 7.7 / 100
- Share of loans on watchlist (rule-based tier): 48.2%
- Overall default rate in the synthetic tape: 5.6%

Office and hospitality assets carry materially higher risk than the rest of the book (9.8% and 18.8% default rates respectively, versus 1.5%–4.0% for multifamily, industrial, and mixed-use), concentrated refinancing exposure sits in loans within two years of maturity, and a moderate macro stress scenario (+100bps floating-rate shock, –10% occupancy, –10% NOI, –10% appraised value) roughly triples the loan volume sitting above an 80% LTV refinancing threshold. Full detail follows.

2. Business Problem & Approach

A CRE lender managing hundreds of loans across office, multifamily, retail, industrial, hospitality, and mixed-use properties needs a continuous answer to five questions: which loans are becoming vulnerable, which borrowers need immediate review, where refinancing risk is concentrated, how rate/occupancy/NOI shocks would move the book, and which assets belong on a watchlist.

Most lenders answer these questions with static, point-in-time borrower financial reviews — typically refreshed quarterly or annually. CRE Sentinel replaces that with a continuously scored, stress-tested view of the whole portfolio, built on four layers:

- A documented synthetic loan database standing in for a real, NDA-protected loan tape
- A credit-risk model producing a 0–100 risk score per loan, with SHAP explainability
- An interactive Streamlit application for live stress testing
- A Power BI dashboard for executive and credit-committee reporting

3. Data & Methodology

3.1 Data

The loan tape (1,000 loans) is synthetic — generated to match realistic CRE underwriting distributions rather than sourced from any real lender, because real loan tapes are protected by non-disclosure agreements. Loan-level ratios (LTV, DSCR, debt yield, interest coverage) are calculated from generated inputs rather than independently randomized, so they remain internally consistent. Default outcomes are drawn from a logistic function of those same ratios, which is what gives a model trained on the data something genuine to learn. Full generation logic is in `scripts/generate_loan_tape.py`; full field-level provenance is in the Data Dictionary.

3.2 Modeling

Two models were trained and compared honestly rather than optimized for a single headline metric: a logistic regression baseline and an XGBoost classifier. Class imbalance (5.6% default rate) was handled via class weighting rather than synthetic oversampling (SMOTE), which risks inflating validation performance on a dataset this size. Three fields — covenant status, delinquency status, and watchlist rating — were deliberately excluded from the feature set because they are generated from the same underlying risk signal as the default label itself, and including them would constitute data leakage.

3.3 Held-out test set results (250 loans)

Metric	Value
Logistic Regression — ROC-AUC	0.788
Logistic Regression — PR-AUC	0.328
Logistic Regression — Recall @ 0.5	0.643
XGBoost — ROC-AUC	0.739
XGBoost — PR-AUC	0.285
XGBoost — Recall @ 0.5	0.286

On this dataset, logistic regression outperforms XGBoost on PR-AUC and recall — a known small-sample effect (only 42 positive examples in training). XGBoost is retained as the production scoring model for its SHAP explainability integration and expected performance advantage as the loan tape scales, not because its metrics are currently better. This trade-off is documented in full in the Model Card rather than omitted.

4. What Drives the Model's Predictions

SHAP (SHapley Additive exPlanations) analysis on the XGBoost model shows loan-to-value, occupancy, and debt yield as the three dominant risk drivers, with interest rate and loan age as secondary factors — directionally consistent with how the underlying default labels were generated, which is a useful sanity check that the model learned real relationships rather than noise.

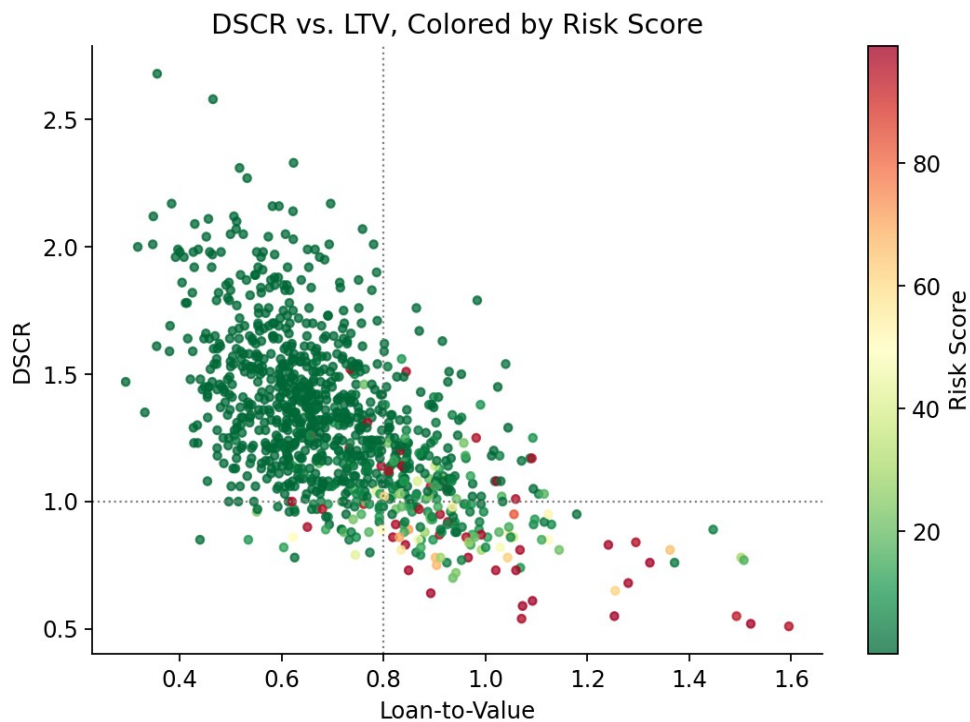


Figure 1. DSCR vs. LTV across the portfolio, colored by model risk score. High-risk loans cluster below 1.0x DSCR and above 80% LTV, as expected.

5. Portfolio Risk Findings

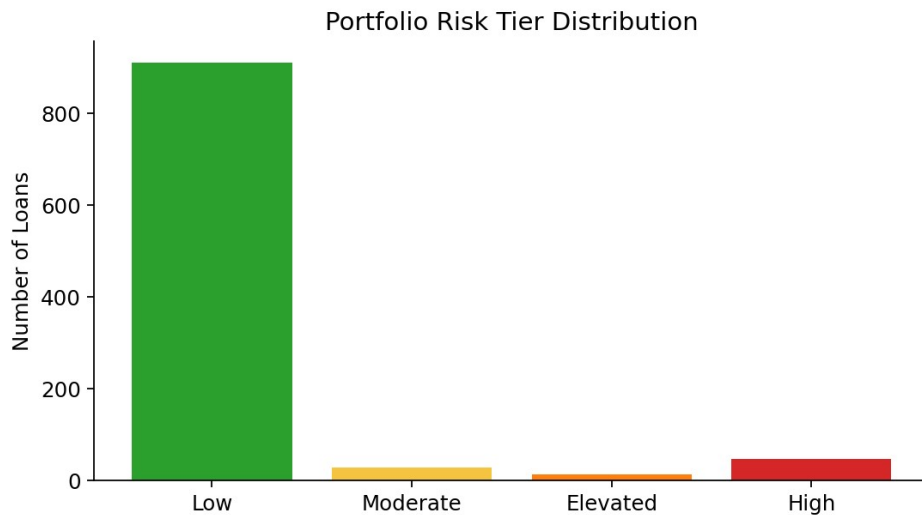


Figure 2. Risk tier distribution across the 1,000-loan portfolio. 46 loans (4.6%) sit in the High tier.

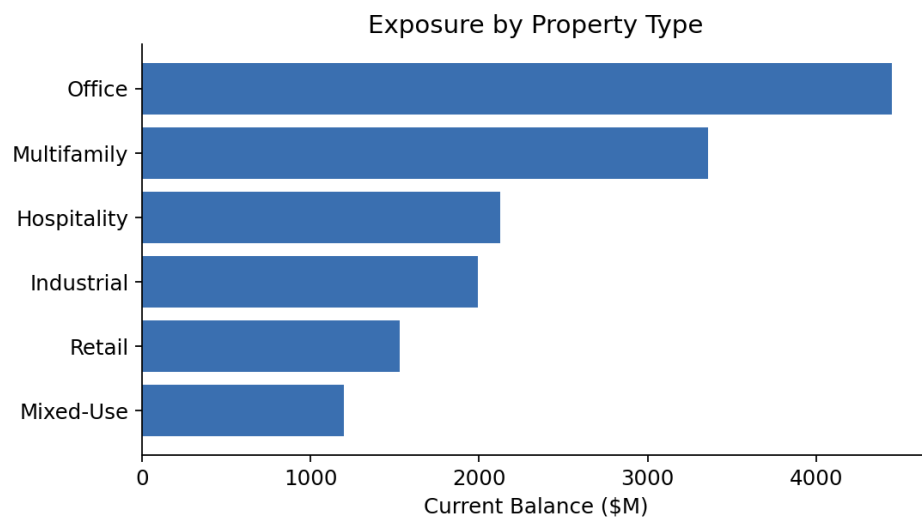


Figure 3. Current balance exposure by property type. Office is the largest single exposure at \$3.4B.

5.1 Default rate and average risk score by property type

Property Type	Default Rate / Avg Risk Score
Hospitality	18.8% / 21.9
Office	9.8% / 14.4
Retail	4.0% / 5.4
Industrial	3.8% / 5.0
Mixed-Use	1.3% / 4.7
Multifamily	1.5% / 2.0

Hospitality and office are the clear risk concentrations in this book, consistent with the broader post-2023 CRE market — office is challenged by structurally lower occupancy, hospitality by cash flow volatility. Multifamily and industrial remain the safest exposures, both in realized defaults and model risk scores.

6. Maturity Wall & Stress Test Results

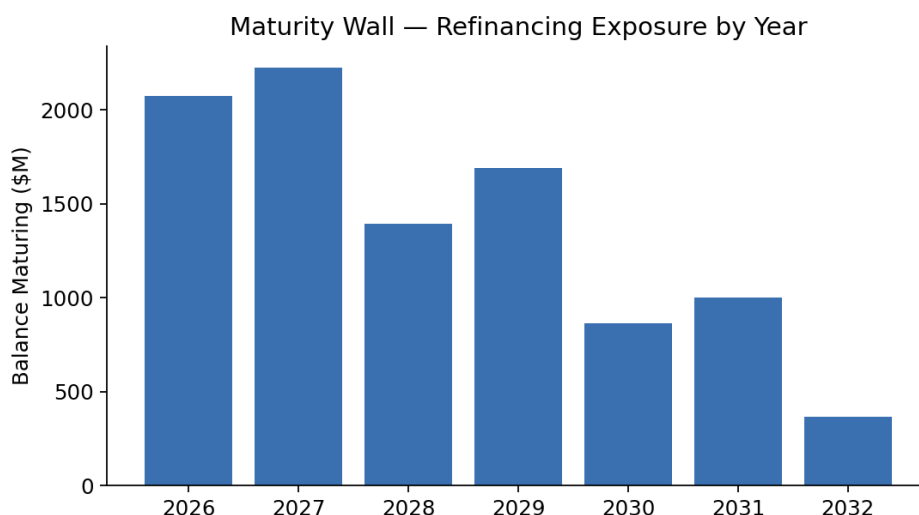


Figure 4. Refinancing exposure by maturity year, 2026–2032.

6.1 Moderate stress scenario

Scenario applied: +100bps rate shock (floating-rate loans only), –10% occupancy, –10% additional NOI shock, –10% appraised value.

Metric	Baseline → Shocked
Average portfolio risk score	7.7 → 12.1
Loans in High-risk tier	46 → 46 (severity rises within tiers before tier migration)
Refinancing gap (loans with LTV > 80%)	\$4,793M → \$7,222M

Even a moderate, realistic macro scenario increases the dollar volume of loans exposed to a refinancing gap by roughly 51%. This is the core value of the stress-testing layer: the baseline snapshot alone understates how quickly refinancing risk can move under plausible market conditions, which is exactly the kind of forward-looking view a static, point-in-time review cannot provide.

7. Model Limitations & Governance Recommendations

This system is built to prioritize which loans a human credit officer reviews first — not to approve, decline, or reprice a loan on its own. Specific limitations:

- False positives: a temporarily depressed DSCR from a delayed reporting deadline or a seasonal occupancy dip can trigger a high risk score for a fundamentally healthy loan.
- False negatives: a distressed borrower temporarily injecting outside capital can mask a failing DSCR right before a real default — confirmed as a measured risk in this project's own confusion matrices, not just a theoretical concern.
- Qualitative blind spots: sponsor reputation, pending lease negotiations, and local zoning changes are invisible to a model trained on financial ratios alone.
- Sample size: 1,000 loans and 56 defaults is a small dataset for this feature count; performance estimates should be treated as directional, not final, until validated on a larger real book.

Recommendation: deploy this system as a triage and monitoring layer that ranks loans for human underwriter attention, with the Credit Review Copilot restricted — architecturally, not just by policy — from taking any credit action. Full detail in the Model Card.

8. Conclusion & Deliverables

This project demonstrates an end-to-end CRE credit-risk surveillance workflow: synthetic data generation with documented provenance, two competing models evaluated honestly rather than cherry-picked, SHAP-based explainability, an interactive stress-testing application, and governance-aware documentation of where the model should and should not be trusted.

Full deliverable set:

- Live Streamlit risk application
- Power BI portfolio dashboard
- GitHub repository with documented pipelines
- Data Dictionary
- Model Card
- This Credit-Risk Report
- AI Prompt & Evaluation Log
- Demonstration video
- LinkedIn carousel